A Modular Framework for Multi-Robot Adaptive Ocean Monitoring *with Gaussian-Process Residual Correction*

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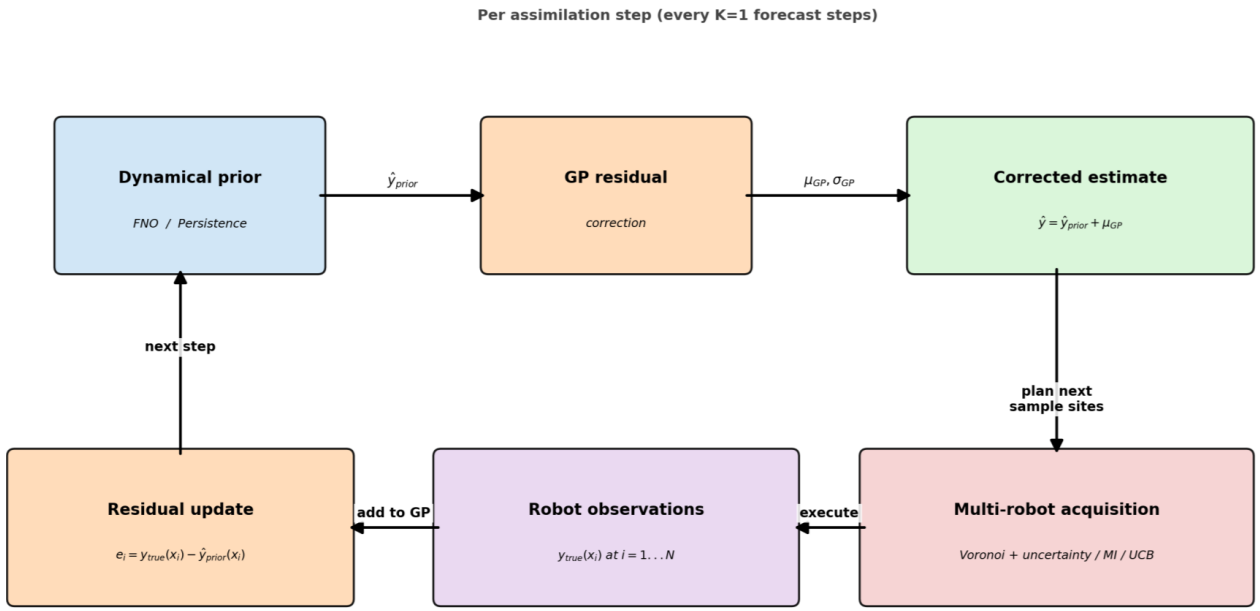
The Problem

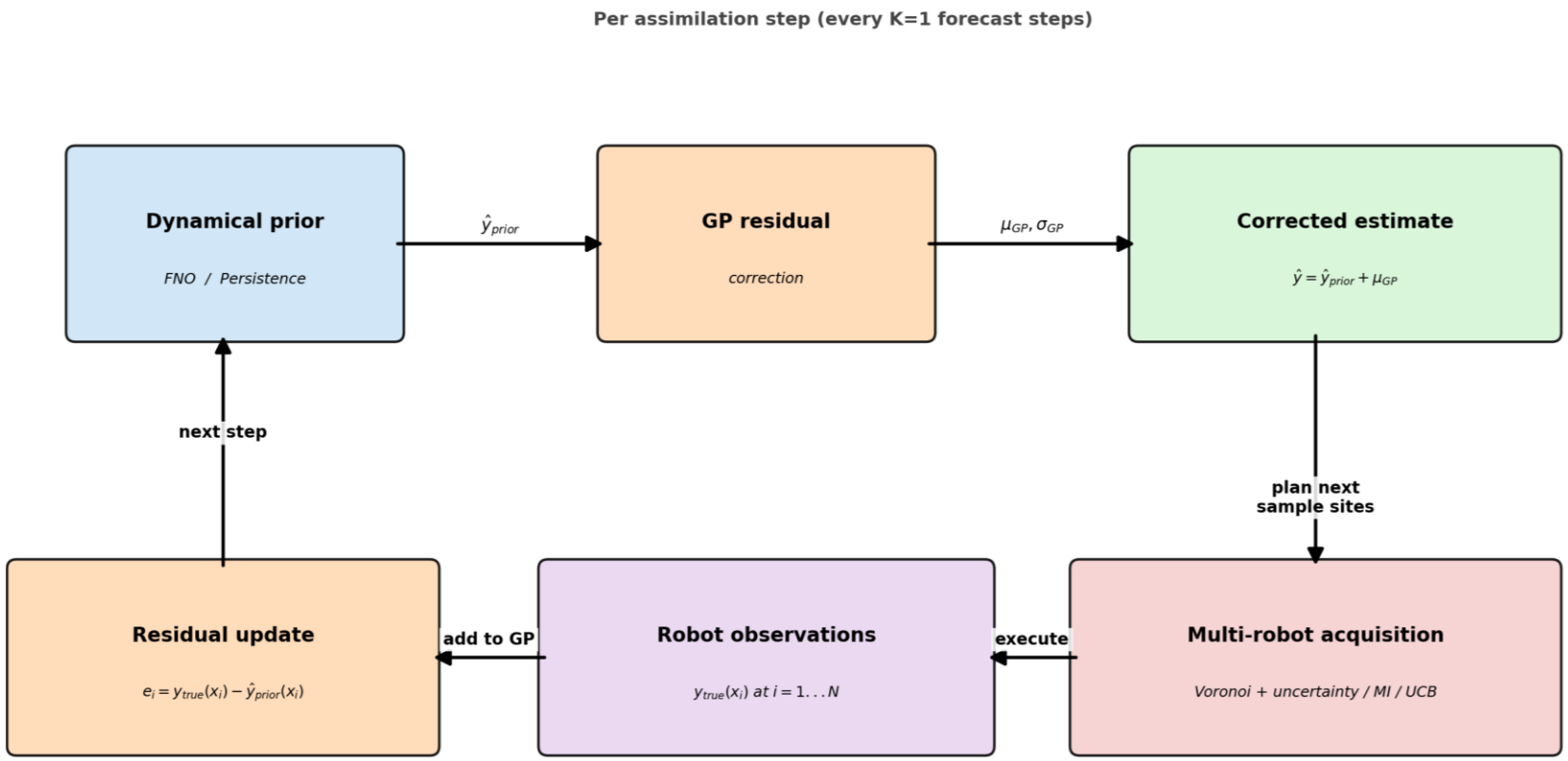
Multi-robot ocean monitoring needs both forecasts and adaptive observations

- Underwater gliders sample sub-kilometer ocean features in situ
- But a handful of platforms must cover ocean basins spanning thousands of km
- Two research communities have grown around this problem — separately:
  - Classical IPP: a GP over the field, no physics
  - Learned forecasts (FNO, OceanNet): basin-scale, no in-situ data
- Gap: no framework integrates a learned prior with multi-robot adaptive sampling



A Modular Closed-Loop Framework

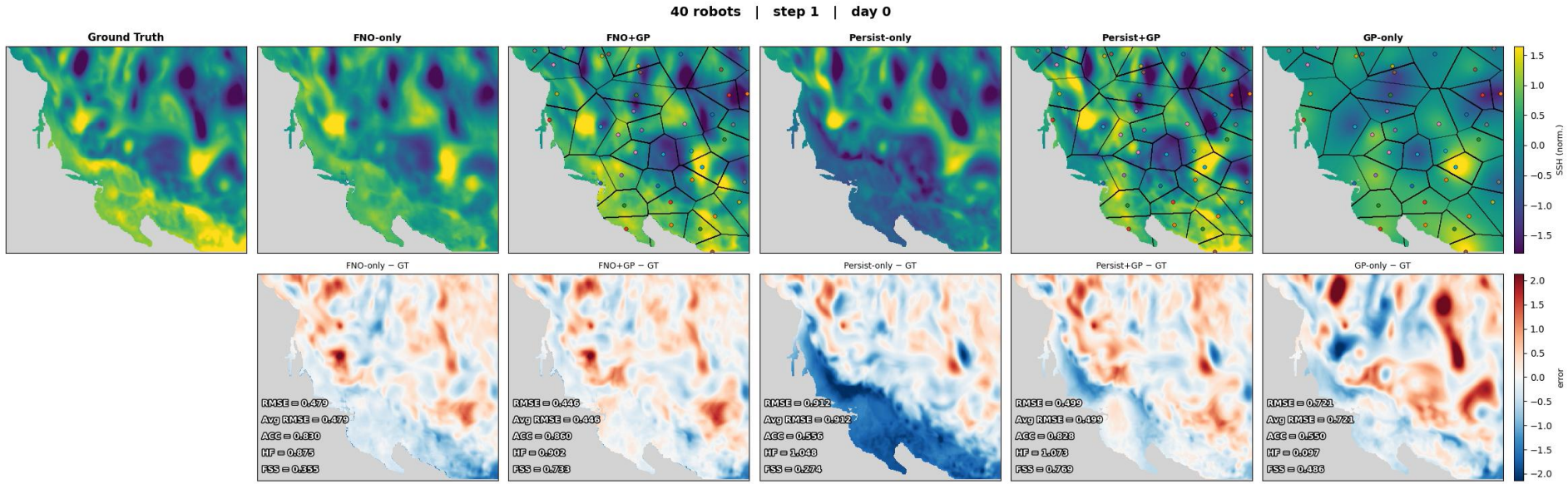
Any dynamical prior + GP residual correction + multi-robot Voronoi acquisition



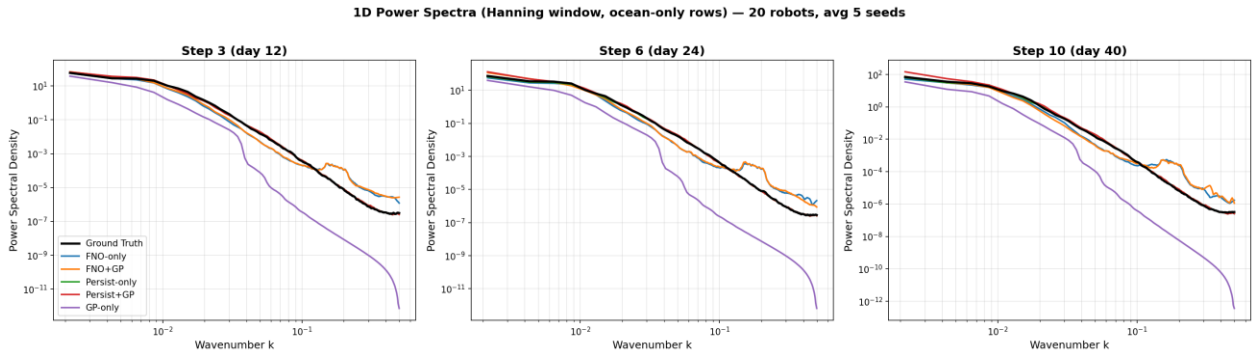
- Prior ŷ_prior = f(ŷ)
- GP fits residual e = y_true − ŷ_prior
- Voronoi acquisition picks next sites
- Robots execute (0.5 m/s glider speed)
- Corrected: ŷ = ŷ_prior + μ_GP
- Same code runs FNO, persistence,
- or no-prior — f is a black box.

RMSE Is Systematically Misleading

A no-prior GP wins RMSE — by predicting the mean



Qualitative: GP-only is a smooth blob

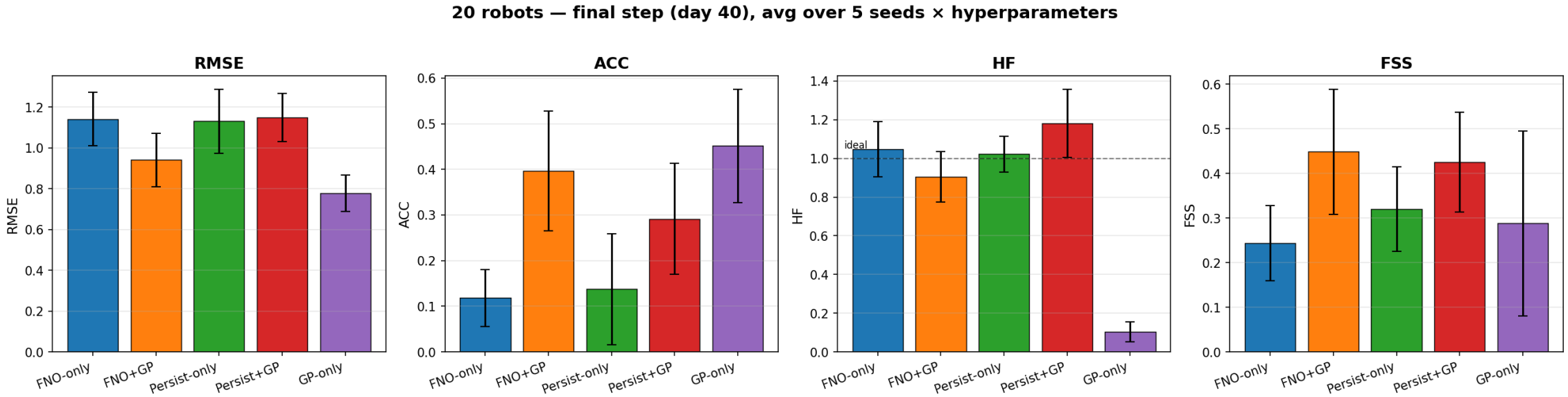


PSD: GP-only collapses 4–5 orders of magnitude at high k

- We adopt 1D Hanning-windowed PSD (standard SSH protocol) and structural metrics (ACC, FSS) as the proper evaluation tools for IPP.

Per-Method Comparison

Read RMSE alone: GP-only wins. Read HF alone: FNO+GP wins.

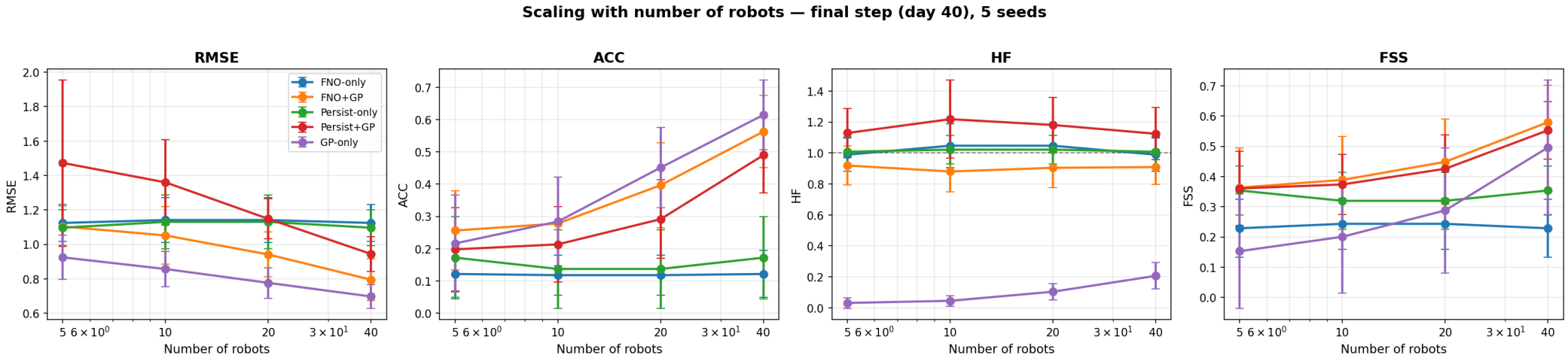


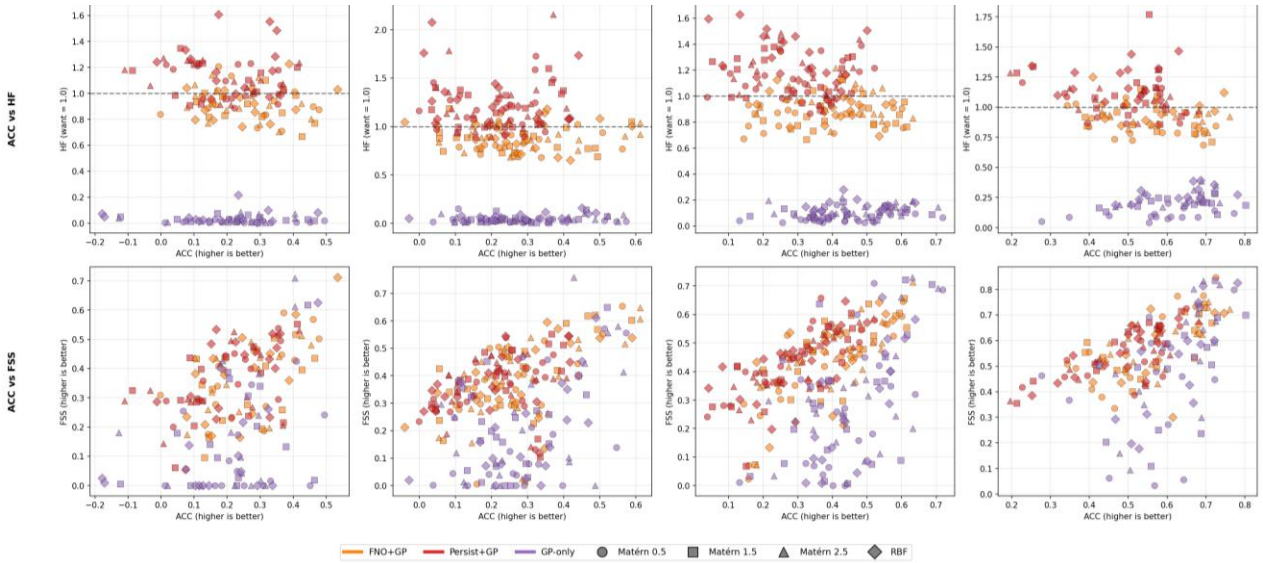
n = 20 robots, day 40

- GP-only: RMSE 0.78, HF 0.10
- FNO+GP: RMSE 0.94, HF 0.90
- Persist+GP: RMSE 1.15, HF 0.85
- GP-only has the lowest RMSE
- but ≈0 high-frequency content.
- Only the joint reading is faithful.

Robustness and Scaling

4 kernels × 3 acquisitions × 4 robot counts × 5 seeds





- Methods occupy distinct metric regions; rankings robust to kernel and acquisition
- Framework benefit grows with observation density; minimum useful regime ≈ 10 robots

Take-Aways

Three contributions of this work

(C1) A modular closed-loop framework

Any prior + GP residual correction + multi-robot Voronoi acquisition. The same code runs with FNO, persistence, or no-prior.

(C2) A spectral evaluation methodology for IPP

1D Hanning-windowed PSD on ocean-only transects + structural metrics. Exposes RMSE failure modes in IPP.

(C3) Empirical analysis across the full hyperparameter sweep

GP loop helps any prior; learned priors yield consistently better structural metrics; no-prior methods cannot recover fine-scale ocean structure regardless of tuning.

Future Work and Acknowledgments

Future Work

- Multi-year and multi-variable generalization (beyond NW Pacific SSH 2020)
- Hardware-in-the-loop validation with real glider fleets
- Multi-objective acquisition for feature tracking (eddies, fronts, blooms)
- Stronger priors in the sparse regime ($n \leq 5$ robots)

Acknowledgments

OceanNet authors for releasing model weights.